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Knowledge boundary spanning and productivity in information systems support community

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ABSTRACT

An information systems (IS) support community represents a knowledge-intensive network where IS professionals interact with end-users to resolve system use problems during the IS post-implementation stage. For IS professionals in the support function, providing support for multiple information systems to multiple business units requires knowledge about different domains (business units or technical systems). In this paper, we take a network perspective to empirically evaluate the effects of an IS worker's network position and knowledge boundary spanning on productivity. Drawing upon theories of experiential learning and knowledge boundary, we perform social network analysis and linear mixed effects modeling to analyze archival data comprising 36 IS workers and 23,450 support requests made by 4568 end-users during the first 13 months post SAP/R3 implementation in a large U.S. organization. Our findings reveal that IS workers' network centrality and boundary spanning positively influence productivity. Surprisingly, their boundary-spanning experience plays a substantially more important role than network centrality. This study makes important contributions to theory and practice in individual experiential learning in knowledge-intensive networks.

1. Introduction

Organizational support for a newly-installed information system (IS) is a knowledge-intensive setting where IS professionals engage in frequent knowledge sharing activities with their business colleagues and provide them with information, knowledge and solutions in a timely manner. Such an IS support community comprises of both end-users (business employees) who use the system to accomplish their work and IS professionals (support personnel) who provide support for system use and maintenance. We consider IS professionals in such a context knowledge workers, as they utilize their knowledge and expertise to support organizational use of the implemented systems. As such, their ability to share knowledge effectively with business employees in an organization is critical to the efficiency of IS support operation. For example, prompt and effective resolutions by IS support personnel enable employees to perform their IS-enabled work, promoting organizational use of IS [7] and facilitating the extended use of IS [24]. In contrast, their failure to efficiently address users' requests is found to cause user frustration and disruption in work [44]. Moreover, the knowledge required to efficiently perform IS support work cannot simply be a body of knowledge to be learned once. Rather, it is acquired through a dynamic process of learning from work-based activities dedicated to the support tasks at hand. Thus, understanding the factors contributing to IS workers' learning and productivity in the support function becomes an important strategic issue for IS managers and business executives.

One important predictor for a knowledge worker's productivity is experiential learning, learning from individual experience. A worker's cumulative experience provides the worker with an opportunity to become more proficient in individual tasks and in performing established routines and practices [2,42]. As knowledge workers increasingly perform tasks in an interconnected environment, their experiential learning is likely to be affected by the network in which they are embedded. In a network, individuals not only learn through their personal experiences, but also benefit from knowledge accumulated by others [23,42]. In the IS support environment, collective learning and problem-solving among IS workers and users underlies a crucial knowledge sharing process [36,46]. In this regard, IS workers' network positions in the IS support community are likely to influence their access to knowledge sources, subsequently affecting their performance in completing IS support tasks. This suggests that examining experiential learning from a network perspective may provide insights into IS workers' productivity.

Yet, experiential learning in a knowledge network may be constrained by the boundaries of knowledge domains, such as business areas or technical systems. Knowledge boundaries often arise around specialized domains of knowledge and give rise to semantic, syntactic and cognitive differences among individuals from different knowledge domains [8,10]. Moreover, knowledge boundaries cause discontinuities

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in knowledge sharing and integration [5,9], which can become problematic in a knowledge network, hindering the flow of knowledge across boundaries and impeding collaborative work. One strategy to deal with the negative consequence of knowledge boundary is boundary spanning, e.g., participating in two different knowledge domains across the boundary. Boundary spanning allows an individual to build the trust with members of both domains and to pass good practices from one domain to another [36]. More importantly, boundary spanning in a network is found to facilitate diverse knowledge exchange and create opportunities for learning from others in the network [60]. As a result, individuals may gain varied experience associated with different knowledge domains [28,36].

In this paper, we take a network perspective and empirically evaluate the effects of network position and boundary spanning on experiential learning and productivity of IS workers. To achieve our research objectives, we integrate Kolb's [26] experiential learning theory and Nonaka's [34] knowledge creation theory to empirically investigate the performance effect of IS workers' learning in IS support. We consider organizational support for implemented information systems as a knowledge-intensive IS support community where IS workers interact with their business colleagues to share knowledge and resolve system use problems. In particular, we examine the effect of spanning two types of knowledge boundaries—business units and technical systems—and examine IS workers' productivity through their efficiency in completing IS support tasks. The research is conducted in the post-implementation support of a newly-implemented enterprise resourcing planning (ERP) system, SAP/R3, in a large U.S. organization. We perform social network analysis and linear mixed effects modeling to analyze an archival dataset comprising 36 IS workers and 23,450 support tasks completed for 4568 business employees during the first 13 months post the SAP/R3 implementation. Our results demonstrate that both network centrality and knowledge boundary spanning have positive effects on IS workers' experiential learning and productivity, with knowledge boundary spanning playing a substantially more important role than network centrality. Theoretically, our findings extend experiential learning theory by incorporating two significant predictors of individual learning and performance—varied experienced via boundary spanning and network position—and propose a network-based model of experiential learning.

The structure of the paper is as follows. We start with the description of our research context of IS post-implementation support and of multi-site implementations of enterprise systems. This section allows us to explain why this research design enables us to examine individual learning of multiple knowledge domains in a knowledge-intensive network. In the Theoretical background and hypotheses section, we develop hypotheses by drawing on studies on experiential learning, knowledge boundary, and social networks. In the Methods section, we describe our research site and data, and specify statistical models. We then present our data analysis and results in the Data analysis and results section, and discuss our research contributions and practical implications in the Discussion section. In the Concluding remarks, we discuss the implications of the findings and outline directions for future research.

2. Investigative context: IS post-implementation support

The core of IS support work involves understanding problematic incidents of system use, uncovering causes to the problems, creating solutions, and delivering the solutions to end-users in the format of information, knowledge, and corrections/fixes. In the IS support context, support tasks are often assigned to an individual worker; individual experience in resolving end-users' problems thus provides an IS worker with a learning opportunity to become more proficient in the provision of IS support services. In such an environment, IS workers involve in the activities of knowledge sharing—pertaining to both the acquisition and provision of knowledge by individuals [43]—with other participants in the support community. Learning and knowledge sharing are important for IS workers to develop proficiency, speeding up their process of completing problem-solving tasks in the technical support context [25]. Hence, the speed at which a support task is completed becomes an important measure of IS workers' productivity.

IS post-implementation support often entails knowledge transfer between support personnel and business employees; knowledge transfer is evidenced from the process of resolving system use problems. When a user (e.g., a purchasing agent) encountered a problem in using SAP/R3's supply chain module to perform a routine task (e.g. processing employee purchase orders), he would call the support center and describe the problematic incident, detailing the steps he performed on the system and the system error message. This problem description often provided IS workers with information on the business context where a system feature was actually applied. Then after the assigned IS worker diagnosed the problem, he would communicate directly with the employee who initially reported the problem, and guided the employee on how to resolve the problem. Hence, the ticket resolution process reflected the process of knowledge transfer and learning between IS workers and users. Moreover, the knowledge transferred was not only about new system features; it could also involve knowledge regarding business processes. Prior research has considered IS support personnel as important sources of technical knowledge for their business colleagues. For example, Santhanam et al. [46] found knowledge of technical systems flowed from IS personnel to business users, and Pawlowski and Robey [36] viewed IS personnel as an important channel to facilitate the knowledge flow between business units. When users encounter problems in integrating the new technology into their work routines [7,17], the interactions between IS personnel and users, and their knowledge sharing activities in particular, become critical in achieving effective system use. As such, prior research has highlighted IS personnel's problem-solving ability for improving their productivity [16].

IS support community provides us an excellent context to examine individual learning in a knowledge-intensive network due to the presence of multiple types of knowledge domains. At the SAP Support Center, IS workers were often associated with one support team, focusing on one specific technical system (e.g., HR/Payroll Management) used by multiple organizational sites. Under this circumstance, the variety of users could create an opportunity for knowledge sharing between user locations, which facilitated the experiential learning of IS workers. The Support Center manager usually assigned open tickets to those IS workers who were available in each support team. When necessary, the Support Center manager rotated the support personnel among the five systems to meet the fluctuating demands for staffing in different systems. In that regard, IS workers could also accumulate experience with multiple technical applications. To encourage knowledge sharing between IS workers, the SAP Support Center adopted several strategies, including weekly meeting to discuss the frequently reported system use problems and IS workers' solutions, posting on the Intranet "Frequently Asked Questions" (FAQ), and forming cross-team focus groups to support the same business unit on major enhancement requests. Informally, the IS workers, sitting in the same office, often consulted with their peers regarding helping the same group of users. Therefore, an IS worker may have access to multiple types of knowledge domains.

3. Theoretical background and hypotheses

Learning at workplace occurs at both individual and collective levels, combining explicit and tacit forms of knowing and theory and practice modes of learning [39]. In developing a framework of network-based experiential learning in the IS support context, we pay attention to an analytical framework which encompasses experiential learning at both individual and network level. We adopt the experiential learning model developed by Kolb [26], which provides a detailed framework capable of capturing different aspects of individual experience, and integrate it with the knowledge creation model by Nonaka [34], which examines experiential learning at the macro level by focusing on the social interactions among individuals.

3.1. IS support community: from individual learning to network-based learning

In post-implementation support, support tasks are often individually performed; therefore, IS workers should benefit from their experiential learning. According to Kolb's [26] classical Theory of Experiential Learning (ELT), learning is defined as "the process whereby knowledge is created through the transformation of experience" (p. 41). Individual experience involves four modes—concrete experience, reflective observation, abstract conceptualization, and active experimentation—through which experience is transformed and knowledge is created. In particular, individuals learn from "hands-on" experience (*concrete experience*) and through trial and error (*active experimentation*). They also learn to evaluate concrete situations from many different perspectives (e.g., brainstorming) (*reflective observation*). In addition, through finding practical uses for abstract ideas, individuals learn to understand a wide range of information and to put it into concise and logical form (*abstract conceptualization*).

Kolb's model provides a detailed explanation for IS workers' learning through individual experience in the IS support context. When a user of the Supplier Relationship Management (SRM) application reports problems in creating a purchase order for medical lab equipment, the IS worker may try to create similar purchase order by applying similar parameters (e.g., item number, vendor ID, quantity, delivery date, delivery destination), thus replicating the problem and learning about the potential causes to the problem (*learning from concrete experience*). The IS worker may change the parameters (e.g., enter different vendor or order quantity) and check if the reported problem still occurs (*learning from active experimentation)*. In addition, the worker may need to follow up with the user to determine his objective and expectation about the purchase order. This allows the worker to evaluate the problematic usage case from different perspectives (*learning from reflective observation*). To complete our analogy, if the worker finds similar problems with purchase order creation (e.g., discrepancy in order quantity or mismatch in vendor ID), the worker learns about the pattern in the purchase order problems and resolution strategies (*learning from abstract conceptualization*).

Individual experiential learning is also likely affected by the interactions among members (other IS workers and users alike) in the IS support community. In this regard, Nonaka's (1994) knowledge creation theory provides good insights. Nonaka explains that knowledge creation centers on the dynamic interaction between two mechanisms—knowledge internalization and externalization—which enable the conversion between explicit knowledge and tacit knowledge. Further, for knowledge creation to be magnified and scaled up from individuals to groups to organizations, two other mechanisms—socialization and combination—are needed. Both mechanisms are enabled through social interactions among individuals. *Socialization* facilitates the tacit-to-tacit knowledge conversion while *combination* facilitates the explicit-to-explicit knowledge conversation. Nonaka stresses that social interactions among individuals play a critical role in the amplification of knowledge; not only a greater amount of knowledge will be generated but also knowledge will be generated at a faster speed as more individuals are involved. Nonaka's theory helps us understand how learning happens at a macro level and in networks.

Next, we integrate Nonaka's [34] knowledge creation model with Kolb's [26] learning model to expand the view of individual-based learning to network-based learning. We specifically examine two important factors—individual's network position and boundary spanning—and develop related hypotheses.

3.2. Network position and experiential learning

Knowledge resides in the experience and expertise of individuals as well as in the collective actions of network participants enabled by their shared communication channels [22,55]. Individuals not only learn through their personal experiences, but also benefit from knowledge accumulated by others in the network [23,42]. The pattern of linkages among these individuals and the relationships built through them not only provide them with opportunities to better identify new knowledge and expertise, but also to serve as channels for mobilizing these knowledge and expertise among them [22,55]. For example, the study of a global virtual community by Chiu et al. [13] found that social interaction ties among members increased the quantity of their knowledge sharing. Prior IS studies have adopted social network approaches in studying new organizational forms in the IS community [47,52].

Similarly, an examination of IS workers' experiential learning would benefit from adopting a network perspective. In the IS support context, IS workers directly work with users to solve their system use problems. The process of analyzing problems and searching for remedies may also involve IS workers and users in discussing the problems and exchanging information. Individuals within a community not only share syntax and semantics from their knowledge domains but also share expectations for the consequences of applying knowledge to a novel situation. For instance, in the new product development project that Carlile [8] studied, engineers from design and manufacturing recognized differences in their functional knowledge and worked collaboratively to comprehend the consequence of applying their knowledge. Likewise, IS workers and users exchange various types of knowledge, ranging from explicit, syntactic knowledge to context-specific knowledge. This process involves the four modes of knowledge conversion between explicit knowledge and tacit knowledge [34]. As a result, an IS worker–user network (Fig. 1a) represents an immediate knowledge-intensive environment.

Social interactions among individuals play a critical role in the process of knowledge sharing and creation via two important mechanisms, socialization and combination [34]. Socialization facilitates the transfer and creation of tacit knowledge through sharing experiences among individuals, while combination involves synthesis and integration of explicit knowledge through social interactions. In the IS support community, IS workers travel to user sites to discuss IS use problems, watch user demos, and identify problem solutions. They also interact with users through emails or phone conversations to discuss and resolve IS use problems. These social interactions promote the interaction between the two modes of externalization and internationalization, amplifying the individual learning and knowledge creation [12]. Thus, socialization can facilitate tacit knowledge exchanges and joint problem solving between IS workers and IS users, especially when the problems are complex and difficult to articulate [34]. Meanwhile, it can also provide an opportunity for IS workers to learn about the routines of coordinating with IS users [42].

Network literature has widely examined the effects of network position on individuals' knowledge transfer and performance. Network centrality has been consistently identified as an important factor (e.g., [22,35]). For example, network centrality in an employee network plays a critical role in improving an employee's knowledge sharing and performance [43]. Likewise, an IS worker with high centrality in the IS support community is likely to accumulate rich repertoire of knowledge and experiences about different users' problems and related problem solutions through interacting with many users. The socialization through direct communication and information sharing with users can speed up the learning process of the IS worker. Consequently, a centrally positioned IS worker is also likely to develop efficient routines of working with different users, improving his productivity.

Hypothesis H1a. Network centrality of an IS worker in the IS worker–user network is positively related to the individual's productivity.

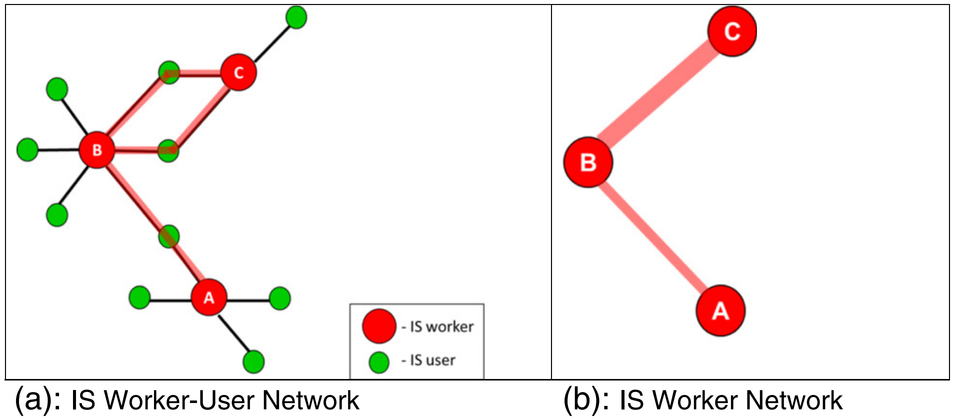

Fig. 1. IS support community. (Red nodes are IS workers; green nodes are IS users; the links between two nodes indicate knowledge flows).

Furthermore, knowledge spillover occurs when the knowledge of a third unconnected person is passed on to an individual via another person. Knowledge spillover has been found to affect the performance of networked R&D firms [35]. In the IS support community, knowledge spillover may often reside in the form of explicit knowledge, i.e., the accumulated knowledge, best practices, and work routines of another IS worker. The knowledge may be passed onto an IS worker through interacting with (and solving problems of) the same IS users. Thus, the network position of a highly central IS worker will not only enable the focal individual to have fast access to directly connected users, but also provide channels through which other indirectly connected users can be reached by the focal individual. In this regard, accumulated knowledge of related problem solutions and efficient work routines of other IS workers may be passed onto the focal individual [20]. Meanwhile, combination and socialization [34] may provide critical mechanisms for the IS worker to subsequently synthesize and integrate knowledge to derive new solutions for those system use problems at hand.

In the IS support, sometimes the same user may encounter multiple problems and may work with different IS workers to resolve those problems. Thus, those IS support workers may share the same sets of users by handling their system use problems. As a result, an IS worker network (Fig. 1b) may represent a secondary knowledge-intensive environment where IS workers may indirectly link to each other through sharing the same sets of users. Through interacting with the shared users, IS workers may benefit from the knowledge spillover of other support workers and learn about the accumulated knowledge, best practices, and routines of problem solutions of other workers [25]. As a result, the individual worker is likely to learn faster about problem resolutions and experience an increased productivity. Therefore, we hypothesize:

Hypothesis H1b. Network centrality of an IS worker in the IS worker network is positively related to the individual's productivity.

3.3. Knowledge boundary spanning and experiential learning

In addition to network position, an individual's performance may also benefit from his experiential learning. Numerous studies in group and organizational learning have documented the link between cumulative experience and measures of performance improvement in different contexts, including manufacturing [2], surgical operations [38,42], service organizations [15], and IS development and maintenance [6,25]. Prior studies have provided several explanations on how people learn from individual experience. One explanation is that individuals learn from a trial-and-error mechanism, which is made possible by repeating a task multiple times [15]. As an individual accumulates experience with practicing the same routines multiple times, he is able to reduce errors and complete the routine faster.

Unlike manufacturing tasks which are structured and repetitive [2], an IS support task is often unstructured and dynamic. Performing an IS support task relies not only on an IS worker's knowledge accumulated from his direct experience but also on his problem-solving ability to create efficient solutions. In this regard, the type of accumulated experience which enhances an individual's capability will be critical for individual learning. Individuals' varied experience (with products or users) is the type of experience that has been found to enhance individual learning capability, especially in knowledge-intensive tasks [49]. In ERP post-implementation support, IS workers encounter two types of knowledge boundaries: (1) business boundaries resulted from different business units, and (2) technology boundaries resulted from different technical applications. As business units within an organization develop their own practices of using a new technical system, they form their communities of system use, which embrace their own local, idiosyncratic routines [57]. When IS workers assist users from those different business units, they find themselves playing a bridging role, passing good practices of system usage from one unit to another [46]. Moreover, the applications (e.g., SRM and HR/Payroll) are built to enable the processes and data of different business functions, thus causing differences in the technical features and functionalities and giving rise to technology boundaries. Hence, in this study, we focus on these two types of knowledge boundaries: business boundary and technology boundary (Fig. 2).

Spanning business boundaries is likely to facilitate IS workers' learning through varied experience. First, business users in one unit may experience context-specific system use that differ from those in another

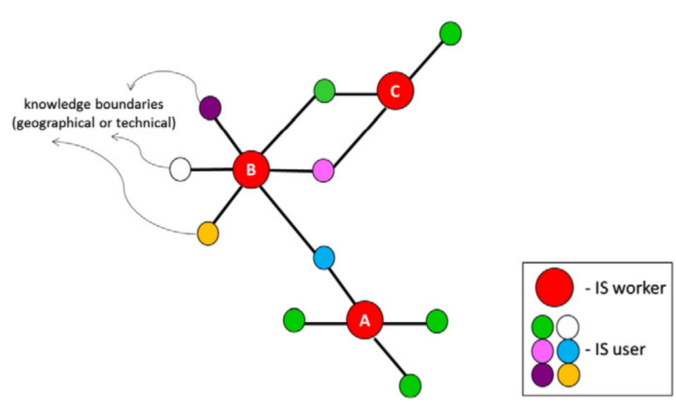

Fig. 2. Varied experience through boundary spanning in IS support community.

unit. The differences in these system use problems require shared cognitive understanding of the contexts. When IS workers interact with diverse users from different business units and locations, they are likely to gain varied experience: (1) through learning different users' problems that are embedded in diverse geographical contexts and work routines; and (2) through learning to coordinate and work with users in diverse contexts [42]. As IS workers accumulate varied experience through spanning the business boundaries, they are likely to become more skillful at coordinating with users in joint problem solving, and more proficient in applying knowledge and solution learned from one context to another.

When the IS workers engage in multiple business domains, individual learning via varied experience can be associated in particular with two learning modes in Kolb's model—reflective observation and abstract conceptualization—although concrete experience and active experimentation are beneficial. When an IS worker assists two users from two different business units (e.g., academic department and emergency room (ER) unit) with similar problems (e.g., purchase order processing), the worker may determine differences in users' objective and expectation about their purchase orders. For example, a purchase order by an academic department usually follows the cycle of academic calendar and is characterized by large quantities. By contrast, purchase orders by an ER unit often are made in short notice and in small quantities. *Learning from reflecting on the differences* in IS support needs allows the workers to assess the urgency of a support task for the SRM system usage and anticipate the consequence and impact of their task completion, leading to an efficient performance. Moreover, interacting with different business units provides workers the opportunities to identify patterns emerging from similar system usage problems, i.e., purchase order processing problems are caused by discrepancies in order prices more frequently in academic departments while by discrepancies in order quantities in ER units. Uncovering these patterns allows a worker to develop strategies for performing similar support tasks in the future, facilitating the individual worker's *learning from abstract conceptualization*.

Moreover, this varied experience associated with multiple geographically dispersed user groups allowed IS workers to better understand problems and difficulties that users encountered with the installed technologies, enabling them to resolve users' problems more efficiently. Prior studies of experiential learning [49] also suggest that the accumulation of varied experience can be critical for developing new capabilities, especially the absorptive capability that enables an individual to evaluate and utilize knowledge from external sources. Similarly, experience with multiple units at different locations not only provides IS workers with opportunities to understand the similarity and difference in users' system usage behavior and information needs, but also allows them to share best use practices across user groups [36]. For the IS workers themselves, as result of learning across diverse business contexts and practices, they solidify their content knowledge and improve their absorptive capacity in learning and problem solving, which enables them to resolve users' problems more efficiently.

Hypothesis H2. The extent an IS worker's spanning business boundary is positively related to the individual's productivity in the IS support community.

Like spanning business boundaries, IS workers also span technology boundaries when they are assigned to support business colleagues using multiple applications. An individual's experience with multiple technical systems may expose the individual to various sources of information and knowledge embedded in the technical systems, allowing the worker to better meet users' needs with the technology. For example, the more experience a developer has with regard to related systems, the better his group will perform in resolving software modification requests [6]. Similarly, IS professionals who are able to transfer knowledge from one problem type (i.e., Operating Systems configuration and installation) to another (i.e., Network registration and routing) can perform better in IS support operations [25]. The variation in experience prevents possible "learning myopia" and "competency traps" [30]. For example, Lapre and Tsikriktsis' study on the airline industry [27] found that diverse experience with different types of aircrafts and route offerings enhances an airline's learning rate in customer service.

In the IS support context, when IS workers participate in the support for multiple technical systems, their learning through the multiple technical domains could be associated with two particular learning modes—active experimentation and abstract conceptualization—in Kolb's model. When an IS worker performs support tasks in relation to two different applications (e.g., SRM vs. HR/Payroll), they learn about the different features and functionalities built in the technical artifacts. At first, solving a SRM system use problem may require different approaches from those used in solving a HR/Payroll system use problem. IS workers may need to rely on "trial-and-error" approaches to evaluate of consequences of different parameters (e.g., user role, field input) to evaluate and address a system use problem. In other words, hands-on experience with the technical applications (*learning from active experimentation*) allows an IS worker to get to know the nuts and bolts of a technical system, enhancing their problem solving work. Yet, ERP system is integrated and process-oriented, i.e., all the applications rely on one central database and follow pre-specified workflows such that a data input and process in the upstream of the system, i.e., grant account in the Grant Management module is affected by the expenditure on a purchase order in the SRM application. In this regard, as IS workers accumulates experience with resolving different but related system use problems, they learn about the patterns and linkages between the two technical domains (*learning from abstract conceptualization*), which allow them to develop strategies for efficiently performing support tasks.

The varied experience with multiple technical applications may enhance an individual worker's ability to assimilate or process acquired information and knowledge to a new and different problem domain. When individuals have the ability to transfer knowledge across domains, it becomes possible for them to create analogous solutions to a new problem domain [32]. As such, an increase in varied experience through technology boundary spanning is likely to result in better performance of an IS worker.

Hypothesis H3. The extent of an IS worker's spanning technology boundary is positively related to the individual's productivity in the IS support community.

In sum, we predict that both network centrality and knowledge boundary spanning affect IS workers' learning and productivity in the IS support community. The figure below (Fig. 3) summarizes all the hypotheses.

4. Methods

4.1. Research site and data collection

The research site is a large U.S. enterprise (referred to as "Organization") that provides patient care, education, and research. The enterprise

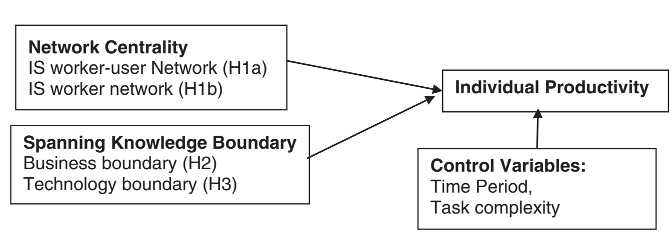

Fig. 3. Network position and boundary spanning on productivity in IS support community.

employed 40,000 people and comprised a health system (Hospital) and one academic institution (University) located in a large metropolitan area in the northeastern region of the United States. To streamline its business functions across its hospital and academic units, the organization adopted the enterprise resource planning (ERP) system, SAP/ R3, and installed five modules, including human resource and payroll management (HR/Payroll), supply chain management, financial management, business warehouse, and project management. The organization required all their employees to utilize the new applications to perform their work ranging from payroll processing to equipment purchase. To provide centralized support for the newly-implemented SAP/R3 system, the organization set up a SAP Support Center staffed with 36 full-time IS workers to assist employees with their requests for information, knowledge and solutions. Each request became a support task for the IS workers. To manage its operation, the SAP Support Center used a ticketing database to record all the user-reported problems. Each ticket record contains data on the sequence of activities in solving a system use problem, from the problem origin, to its categorization and assignment, and to the final resolution of the problem. The ticket logs in the database enabled us to construct a comprehensive and detailed network of the IS support community.

We extracted archival records of a total of 23,450 support tasks, related to the five applications (modules) made by 4568 end-users from six major organizational sites in geographically dispersed locations over a 13-month period (04/2007-04/2008) to allow us to examine the learning effects. Accordingly, we obtained information about who (IS worker) resolved which ticket from whom (IS user), and assigned a tie between them. In total, 13 monthly IS worker–user networks were constructed. Fig. 4 depicts a one-time snapshot of the network in April 2007.

As shown in Fig. 4, IS support community can be viewed as a knowledge-intensive network, where IS workers interacted with IS users to solve system use problems. Each tie was treated as unidirectional as the knowledge and information related to a system use problem can flow in both directions. We consider a support task as the basic unit of activity in the IS support, as suggested by our field data and interviews. As IS workers were randomly rotated across business units and technical systems, they did not always have dense links (connections) in the IS worker–user network.

To capture knowledge spillover effects, we further constructed 13 monthly IS worker networks based on IS worker–user networks. Fig. 5 presents one time snapshot of an IS worker network in April 2007, which depicts how IS workers were interconnected through sharing the same sets of IS users.

4.2. Variable measures

4.2.1. Productivity

We measured an IS worker's productivity as the *Average Effort* (AvgEff) spent on resolving system use problems. As the labor cost of IS support personnel accounts for the largest component of the cost in IS support service [1], we use an IS worker's resolution time per ticket to measure his productivity, consistent with prior studies (e.g., [16]). The average resolution time is an important measure for customer service quality because prompt responses to customers' problems would increase customers' satisfaction and minimize disruptions on customers' routine work. In IS support service, an individual may be assigned multiple support tasks during the same time period. Therefore, we adopted the approaches used by prior IS studies [6,33] to compute average resolution time per task, as detailed in the Appendix. This

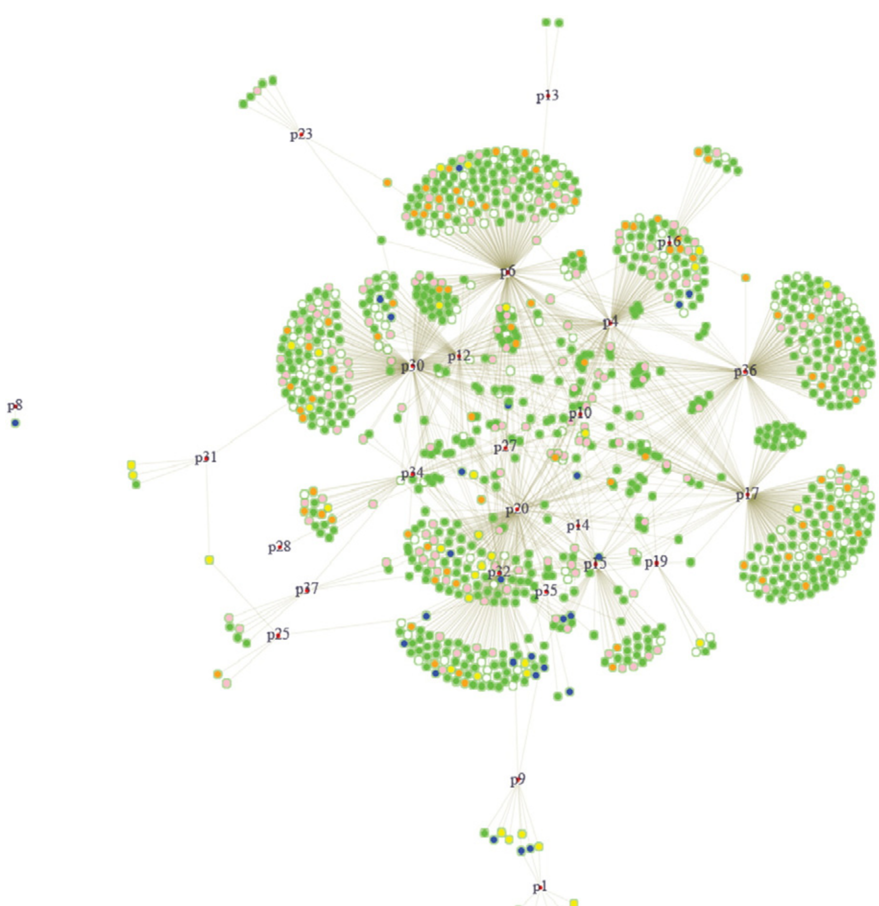

Fig. 4. IS worker–user network in April 2007. (IS workers are indicated by red nodes with number labels; nodes in blue, green, orange, pink, white, and yellow represent users from six different geographical sites.)

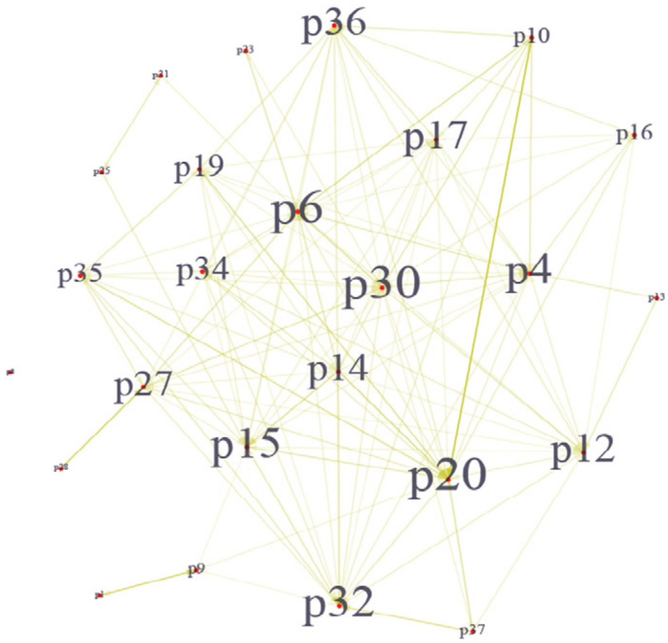

Fig. 5. IS Worker Network in April 2007. (IS support workers benefit from the other workers' knowledge through sharing the same users.)

computation of effort took into account the possibility that (1) an individual may be responsible for multiple tickets during the same time period, and (2) the total number of tickets assigned varies daily.

4.2.2. Network centrality

In this paper, we particularly examined two network centrality measures. (1) *closeness centrality* of IS workers in the IS worker–user network, and (2) *degree centrality* of IS workers in the IS worker network. For the *IS worker–user network*, we focused on the *closeness centrality* because we were interested in studying the network position that captures the knowledge transfer through direct ties and indirect ties between the IS workers and IS users. Closeness centrality provides such a measure that captures an individual's reachability or the average geodesic (shortest) distance to others in the network. A high reachability reflects an individual's capacity to access to a large amount of information in the network by considering both direct ties and indirect ties of the individual. Indirect ties are important to consider as they not only influence an individual's manipulation of the network but also provide channels through which socially distant information and knowledge can reach the individual [20]. For the *IS worker network*, we focused on the *degree centrality* of IS workers because we were interested in examining the extent to which an IS worker was linked to other peer workers through sharing the same users and the knowledge spillover that he may receive through shared users. Knowledge spillover occurs when the knowledge of a third unconnected person is passed onto an individual via another person.

(1) *Closeness centrality* (Closeness1) is calculated using Valente and Foreman's [56] measure of radiality, which is a variation of closeness centrality. This measure takes the average of the reverse geodesic (shortest) distance between two individuals to reflect the extent of connectedness and reachability of an IS worker to others in the network. The closeness centrality of IS worker i is calculated as follows:

$$\text{Closeness1}_i = \frac{\sum_{i \neq j} RD_{ij}}{N-1}$$

where RD_{ji} is the reverse distance computed from the geodesic distance between support person i and individual j. N is the network size of the IS support community. RD_{ji} is computed by finding the geodesic distance and then subtracting the geodesic distance from one plus the maximum value within the geodesic matrix. Each individual's closeness centrality score is then obtained by averaging their reversed distance scores to every other individual. Both the distance values for nodes not reachable to one another and the values on the diagonal are set to zero in the reversed distance matrix [55, p. 92].

(2) *Degree centrality* (Degree2) is calculated by summing up all the direct contacts of an IS worker in the IS worker network [58].

$$\text{Degree2}_i = \sum_{i \neq j} d_{ij}$$

where $d_{ij} = 1$ when IS workers i and j are linked, $d_{ij} = 0$ when i and j are not linked.

4.2.3. Spanning business boundary (ExpLocSpan)

We use Shannon's [50] entropy measure to calculate the diversity of users who are from different business units in the organization and are supported by IS worker i. This measure is used to capture the extent of varied experience through business boundary spanning. A higher score suggests a greater extent of varied experience, while a lower score suggests a smaller extent of varied experience (in other words, a greater extent of specialized experience).

$$\text{ExpLocSpan}_i = -\sum_s p_s (\ln p_s)$$

where s denotes six business sites where users are located, $p_s = n_s/n$ denotes the extent to which users are located at site s and supported by IS worker i; n = total number of users across all sites who are supported by IS worker i; n_s = number of users at site s who are supported by i.

4.2.4. Spanning technology boundary (ExpProdSpan)

Similarly, we use Shannon's [50] entropy measure to calculate the diversity technologies (technical applications) supported by IS support person j. This measure is used to capture the extent of varied experience through technology boundary spanning. A higher score suggests a greater extent of varied experience, while a lower score suggests a smaller extent of varied experience (in other words, a greater extent of specialized experience).

$$\text{ExpProdSpan}_i = -\sum_k p_k (\ln p_k)$$

where $k = 1,2,\ldots 5$ technical applications which the IS use problems are concerned with, $p_k = n_k/n$ denotes the extent to which user problems are concerned with application k that are supported by IS worker i; n = total number of IS use problems across all applications which are supported by IS worker i; n_k = number of user problems that are concerned with application k, which are supported by i.

4.2.5. Control variables

Control variables include prior experience, month, function, and ticket complexity. *Prior experience* (PriorExp) is an adjusted measure reflecting an IS worker's accumulative experience in resolving SAP system use problems prior to the current time period (of month). *Month* refers to the 13 months starting in April 2007; it was added as a control variable in the model because our research focus was on studying the learning effect demonstrated through gained experience by knowledge boundary spanning. *Function* refers to the five teams designated to support the five installed IT applications. The *ticket complexity* refers to the extent to which a large number of components are related in completing a support task, similar to the component complexity proposed by Wood [61] and informed by the insights of the Support Center Manager during our site interviews. The manager explained to us that the tickets remaining open for more than two weeks were complex tickets, and suggested us to use that as proxy for ticket complexity. We thus measure the ticket complexity by calculating and using three standard deviation of the mean ticket resolution time in the data sample as a criteria such that ticket complexity score is 1 if over or below 3 standard deviations else the complexity score is 0. Then the ticket complexity scores are aggregated on a monthly basis per worker. We also controlled for other variables including Gender and Organizational Tenure. None of them was significant and was dropped from the final models (thus not reported here). Table 1 presents the descriptive statistics and correlations of the variables.

5. Data analysis and results

We combined the ticket log data and network measures with individual characteristics of IS workers, and compiled an unbalanced panel dataset of 320 observations involving 36 IS workers over 13 months for data analysis. The dependent variable, IS workers' *Average Effort*, and all other variables (except *Function)*, are continuous variables and demonstrate skewed distributions. Thus, we transformed them with a natural logarithm.

5.1. Linear mixed effects model

We use linear mixed effects models [41,53,62] to test hypotheses, given the variables at individual and network levels. Our model specification below combines the two-level models. Level-1 model of linear mixed effects models represents each IS worker's trajectory of change as a function of person-specific parameters plus random errors. The level-2 model describes the variation in these change parameters across a population of persons. Researchers in the education field often use the linear mixed effects model (also referred to as "hierarchical linear model") to study the effect of school programs and policies on students' individual academic performance, given the nature of multilevel, hierarchical data [40,45]. Linear mixed effects models provide a number of advantages over the other common techniques, such as multivariate repeated measures analysis of variance (MRM) and structural equation modeling (SEM), for modeling individual change [41]. First, linear mixed effects models allow a wide variety of data structures. Unlike conventional MRM and SEM which require that every person have a fixed time-series design (that is, the number and spacing of time points must be invariant across people), linear mixed effects models are generally more flexible in terms of its data requirements because the repeated observations are viewed as nested within the person rather than as the same fixed set for all persons as in MRM. Both the number of observations and the timing of the observations may vary randomly over participants. In addition, linear mixed effects models allow level-1 predictors having random effects to have different distributions across individuals, while MRM and SEM require that the level-1 predictors having the random effects must have the same distribution across all individuals in each subpopulation. Linear mixed effects models also allow a wide array of covariance structures, which makes them more flexible to use.

Model specification:

$$
\begin{aligned}
lnAvgEff_{ti} = {} & \beta_0 + \beta_1 Month_{ti} + \beta_2 Function_{ti} + \beta_3 lnPriorExp_{(t-1)i} \\
& + \beta_4 lnTicketComplexity_{ti} + \beta_5 lnCloseness1_{ti} \\
& + \beta_6 lnDegree2_{ti} + \beta_7 lnExpLocSpan_{ti} (or\ lnExpProdSpan_{ti}) \\
& + e_{ti} + u_i e_{ti} \sim N(0, \sigma_i{}^2) u_i \sim N(0, \tau)
\end{aligned}
$$

where $t = 1,2,\ldots,13$ months, $i = 1,2,\ldots,36$ workers.

The model examines the effect of network centrality (*lnCloseness1* and *lnDegree2*) and the varied experience via boundary spanning (*lnExpLocSpan* or *lnExpProdSpan*) on individual learning and productivity. Each IS worker is estimated with a different residual variance [18,53,62]. We started with a model with homogeneous residual variance across IS workers and then compared it with the model with heterogeneous residual variances across workers. The log likelihood test shows that the model with heterogeous residual variances is signficantly better than the one with homogeous variance. We thus use hetergeneous residual variances for model specification.

We compared various mixed effects models with different covariance structures, including the fixed effects model, the random intercept model, the random intercept and random slope model, and the compound symmetry and AR(1) model. We used maximum likelihood method to estimate parameters and compare models. Results suggest that both the random intercept model and AR(1) model provide the best model fit. Results from both models are consistent with each other. We report random intercept model in the paper and then use the restricted maximum likelihood method [41] to estimate the final model.

5.2. Results

Table 2 presents the results for Models 1–6. Specified as $lnAvgEff_{ti} = \beta_0 + 1\ Month_{ti} + e_{ti} + u_i$, Model 1 examines a mean learning trajectory of IS workers over time. In Model 2, we add in control variables of *Function, lnPriorExp*, and *lnTicketComplexity*. Models 3–4 test the effects of IS workers' network positions in IS worker–user network (*lnCloseness*) and in IS worker network (*lnDegree2*). Models 5–6 test the effects of spanning business boundary (*lnExpLocSpan*) and product boundary spanning (*lnExpProdSpan*) respectively. We conducted robustness check for two final Models 5 and 6. We checked residuals and found no major concerns for auto-correlation and violations of constant variance and normality. We also checked for multicollinearity and found no major concerns. All VIFs range between 1.8 and 8.5.

In Model 1, the estimated $\hat{\beta}_0 = -0.988$ suggests that the mean IS workers' *Average Effort* at the initial month April 2007 was estimated to be 0.372 (i.e., $e^{-0.988}$). The variance estimate $\hat{\tau} = 0.982$ suggests that IS workers differ in their initial *Average Effort* in April 2007 with an estimated standard deviation of 0.991. The estimated $\hat{\beta}_1 = 0.006$ suggests that the mean IS workers' *Average Effort* was estimated to increase at a rate of 1.006 (i.e.,$e^{0.006}$), which was not a significant learning trajectory.

In Model 3, we added the term *lnCloseness1*. The negative significant coefficient (-0.413) of *lnCloseness1* suggests that the IS worker's

Table 1
Descriptive statistics and correlations.

Variable	Mean	Std Dev	Min	Max	1	2	3	4	5	6
1. *AvgEff*	0.827	0.946	0.057	10.000	–					
2. *PriorExp*	30.598	45.354	0.100	332.969	−0.248	–				
3. *TicketComplexity*	0.226	0.277	0.000	1.000	0.682		–			
4. *Closeness1* (IS worker–user network)	17.042	3.766	7.657	25.895	−0.421	−0.011	−0.392	–		
5. *Degree2* (IS worker network)	0.592	0.281	0.000	1.000	−0.649	0.413	−0.651	0.476	–	
6. *ExpLocSpan*	0.156	0.158	0.003	0.591	−0.594	0.573	−0.583	0.373	0.778	–
7. *ExpProdSpan*	0.134	0.128	0.003	0.614	−0.584	0.501	−0.585	0.370	0.773	0.955

N = 320; absolute value of correlation coefficient greater than 0.1 is significant at p < 0.001.

Table 2
Results for the linear mixed effects with a random intercept model.
Dependent Variable: *lnAvgEff.*

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Fixed effects						
Intercept	−0.988*** (0.177)	−0.357 (0.224)	0.901* (0.387)	−0.264 (0.400)	−1.407*** (0.307)	−0.136*** (0.336)
Month	0.006 (0.008)	0.019** (0.006)	0.005 (0.008)	0.027*** (0.007)	0.025*** (0.006)	0.023*** (0.006)
HR/Payroll		1.089*** (0.242)	1.016*** (0.223)	0.764*** (0.093)	0.479*** (0.069)	0.589*** (0.056)
Supply chain		−0.168 (0.229)	−0.176 (0.210)	−0.471*** (0.085)	−0.242*** (0.061)	−0.248*** (0.045)
Security		0.747* (0.331)	0.714* (0.304)	0.575*** (0.146)	0.346** (0.113)	0.496*** (0.104)
Finance		1.047*** (0.243)	1.020*** (0.222)	0.803*** (0.087)	0.461*** (0.066)	0.619*** (0.047)
lnPriorExp		−0.048** (0.016)	−0.064*** (0.017)	−0.075*** (0.015)	−0.017 (0.011)	−0.043*** (0.012)
lnTicketComplexity		0.341*** (0.012)	0.322*** (0.012)	0.293*** (0.011)	0.173*** (0.010)	0.191*** (0.010)
lnCloseness1 (H1a)			−0.413*** (0.109)	−0.097 (0.125)	−0.175† (0.094)	−0.141 (0.102)
lnDegree2 (H1b)				−0.500*** (0.045)	−0.036 (0.046)	−0.113* (0.045)
lnExpLocSpan (H2)					−0.534*** (0.025)	–
lnExpProdSpan (H3)						−0.455*** (0.027)
Random effects						
Residual variance, $\sigma_i^2, i = 36$	…	…	…	…	…	…
Variance of IS person mean, $\hat{\tau}$	0.982	0.075	0.062	0.000	0.000	0.000
AIC	669.64	382.222	372.445	299.054	78.007	143.023
BIC	816.603	551.796	545.788	476.165	258.887	323.902
Deviance (−2 log likelihood)	591.638	292.222	280.445	205.054	−17.993	47.023
Δ (Deviance difference)		299.416 ***	11.777 ***	75.391 ***	223.047*** (vs. Model 4)	158.032*** (vs. Model 4)

N = 320; standard errors in parenthesis. †p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001

closeness centrality in the IS worker–user network has a significant decreasing effect on the IS worker's average effort. Thus, H1a is supported.

In Model 4, we added the term *lnDegree2*. The negative significant coefficient (−0.500) of *lnDegree2* suggests that IS worker's degree centrality in IS worker network has a significant decreasing effect on the IS worker's average effort. Furthermore, at its presence, the effect of network position in IS worker–user network becomes insignificant. Thus, H1b is supported. This result suggests that IS workers' productivity may benefit more from the knowledge spillover of peer workers than from users.

Due to high correlation between *lnExpLocSpan* and *lnExpProdSpan*, we analyzed them in two separate models (Model 5 and Model 6). In Model 5, we added the term *lnExpLocSpan*. The negative significant coefficient (−0.534) of *lnExpLocSpan* suggests that spanning business boundary has a strong decreasing effect on the IS workers' effort. Holding the effect of the boundary spanning constant, the effects of network position on individuals' effort are signficantly reduced. In the IS worker–user network, the performance effect of an IS worker's network position was weakened to −0.175, which was marginally significant at 0.1 level. Meanwhile, the effect of the network position in the IS worker network was weakened to −0.036, not significant at 0.05 level. Thus, H2 is strongly supported.

In Model 6, we added the term *lnExpProdSpan*. The negative significant coefficient (−0.509) of *lnExpProdSpan* suggests that spanning technology boundary has a strong negative effect on the outcome. In the presence of this knowledge boundary spanning, the performance effects of network position are significantly reduced. Specifically, the effect of network position was weakened to −0.141 (not significant) in the IS worker–user network, while the effect in the IS worker network was weakened to −0.113, which was marginally significant at 0.05 level. Thus, H3 is strongly supported.

6. Discussion

This study intended to examine individual experiential learning in a network setting by investigating two influencing factors, knowledge boundary spanning and network position. We found that both types of factors contributed to IS workers' productivity but boundary spanning exerted a more significant and positive impact than network position. First, IS workers benefit from their varied experience with multiple business units; when serving multiple units, an IS worker develops better understanding of users' difficulties with the new technical applications than their less experienced colleagues. This enriched knowledge enables the IS worker to anticipate users' unique system use problems and to address those problems with the most appropriate problem-solving move. It is an important strategy to match the problem-solving moves with problem types in technical support to improve efficiency [16,37]. Moreover, the varied experience enables an IS worker to develop special languages and terminologies used by each unit, further improving the communication and relationship building between IS workers and business units [11].

Further, IS workers' experience with multiple technical applications was a significant predictor of IS workers' productivity. The varied experience affects individual performance probably via two mechanisms. First, those IS workers have a broader scope of knowledge about the functionalities built in different systems than their colleagues specializing in one system or two. The heterogeneity in knowledge domains allows an individual to reflect on concrete experience and to achieve abstract conceptualization, thus enhancing his problem-solving ability in a different system [26]. Second, spanning technology boundaries puts the knowledge (of other systems) accumulated by other IS workers at the disposal of the focal individual. As individuals gain more experience with different technical applications, they are able to apply what they learn from one technical system (i.e., best practices, tools, procedures and tips) to solving problems with another. In this case, knowledge is transferred from one technology domain to another.

Our data analysis also suggested the significance of expanding individual-based learning theory to a network-based theory of experiential learning. The latter highlights that individual experiential learning is enhanced via (1) the varied experience that individual workers gained from engaging in multiple organizational units and technical applications; (2) the network position of the workers which affects the efficiency of knowledge transfer and learning, reaching maximum amount of people and speeding up the knowledge flow from network members to the IS workers. First, our data provided evidence to demonstrate that individual IS workers became more productive as they gained diverse experience about their customers and the technology products. These findings echo those in prior studies highlighting the importance of experience variation on performance [6,49]. However, our study offered further insights on the two important mechanisms to accumulate the varied experience in a network, namely spanning business and technology boundaries. As individual workers learned about the specific elements (business tasks, technical features) in relation to a business unit or a technical system, they combined those concrete experiences with their prior stock of knowledge in developing solutions and creating new knowledge for future use. Kolb's ETL model [26] provides rich details on illustrating the experiential learning at individual level, by distinguishing two types of experience (concrete experience vs. abstract conceptualization) and two types of learning modes (active experimentation vs. reflective observation). In our study context, it is important to understand how this learning occurred, i.e., how the varied experience enhanced individual learning. Individual IS workers may start with actively interacting and communicating with users to learn about their system use problem details before applying their abstract knowledge into creating solutions.

Second, individual performance can benefit from the social interactions in knowledge networks. Nonaka's model [34] identifies modes of socialization and combination that are enabled through social interactions and enables individual learning to be magnified and scaled up to group, organization, and community as a whole. We propose that shifts among the four modes of individual experiences enable the tacit and explicit knowledge conversion during the individual's knowledge internalization and externalization. Socialization and combination provide important mechanisms for speeding up and magnifying the learning and knowledge creation of the network as a whole through social interactions among individuals. As a result, individuals may benefit from an increased learning rate and performance of the network. In addition, network centrality and varied experience through boundary spanning are important factors explaining the difference in individual performance. Therefore, we propose a network-based theory of experiential learning which incorporates knowledge boundary spanning and network position (Fig. 6).

As IS users from different business units may encounter similar system use problems, IS workers who serve multiple units would be able to channel the problem resolutions and lessons learned from one unit to another [36]. Similarly, as shared IT systems embed similar structure and access to same databases, IS workers who support multiple integrated technical systems may apply their learning of one system to another [31]. Our study further suggests that, individual workers' varied experience accumulated from spanning business and technology boundaries in the IS support community enhanced the IS workers' learning and productivity to a larger extent than one's central network position did.

7. Contribution and implications

Our study makes theoretical contributions in two steams of research —experiential learning and IS post-implementation —and provides practical implications on managing IS support organizations. Each of these contributions and implications is discussed below.

7.1. Contribution to research on experiential learning

We believe that incorporating both models—Kolb model [26] and Nonaka model [34]—to explain the complex learning phenomenon in a knowledge-intensive setting such as IS support community is one major contribution to the learning studies. Traditionally, learning from experience has been investigated at a particular level, including organization, group or individual. For example, at organizational levels, numerous studies have documented the link between cumulative experience and measures of performance improvement in different contexts, including manufacturing [2], service organizations [15], and IS developments [6,25]. At group level, the performance benefits of experiential learning have been evidenced on surgical operation teams [38,42] and by student groups [49]. However, learning at the three levels could be related: organization and group learning is, to some degree, a function of the learning of individuals in the group or organization. Simon [51] argues that all intuition, insight, and innovative ideas occur at the individual level but that these ideas are then shared and interpreted within the group. This shared understanding may subsequently become institutionalized in organizational routines or artifacts [2]. By incorporating both Kolb model [26] (focusing on individuals learning by themselves) and Nonaka model [34] (focusing on individual learning in a social context), our paper makes an initial attempt to provide a comprehensive picture of how learning occurs at individual levels and then scaling up to the community through the interactions between IS workers and users in a knowledge network.

In particular, we incorporate varied experienced via boundary spanning and network position, the two significant predictors of individual learning and performance in a network-based model of experiential learning (Fig. 6). Our proposed model has potential to offer insights to enhance individual learning and performance in other knowledge intensive networks such as the network of consulting practice. In the networks of management consultants [14], consulting firms may assign their knowledge workers to multiple client sites and working with

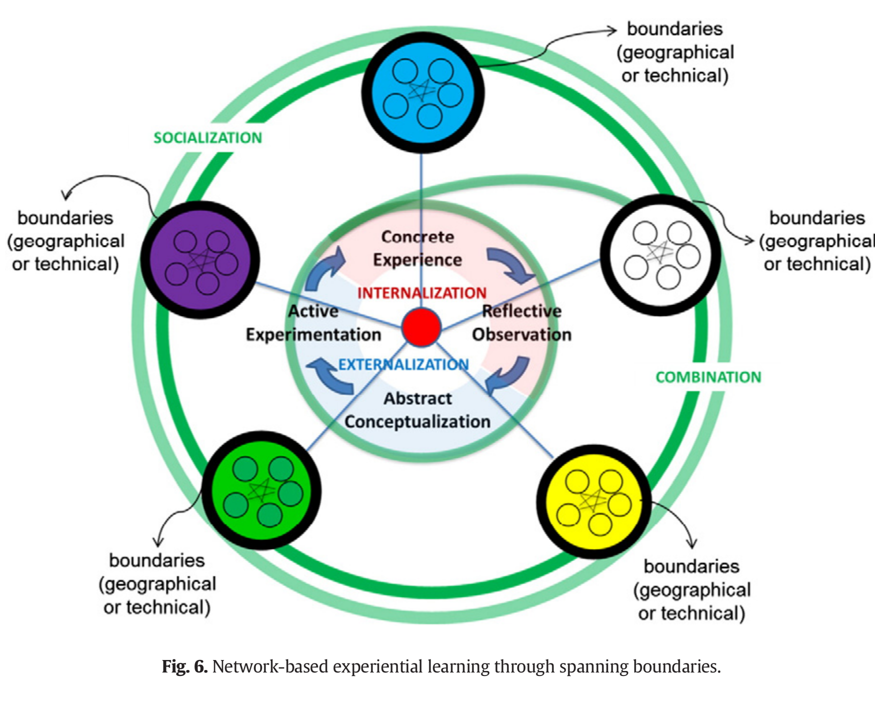

Fig. 6. Network-based experiential learning through spanning boundaries.

multiple projects. The boundary spanning activities performed by consultants and their networking with colleagues are likely to influence their learning and performance.

7.2. Contribution to research on IS post-implementation support

This study makes several theoretical contributions to IS research. First, this study contributes to research on IS support activities by examining IS workers' experiential learning in a network context. Resolving a system use problem involves interactions with IS users and understanding users' behaviors related to information technologies. Thus, the knowledge and skill required of IS workers in accomplishing IS support tasks are more specialized and dynamic, compared to the knowledge and skills required in processing loan application files [54] and in debugging a software defect [33]. Our study provides a contingency view and suggests that balancing the two strategies, specialization versus variation, should carefully consider the knowledge and skill requirements of the task on hand.

Second, this study offers a nuanced theorizing and empirical examination on the individuals' boundary spanning and productivity, contributing to the studies of IS professionals. Prior studies have suggested that spanning user boundaries by IS professionals plays a positive role in IS development [29,48]. However, this literature does not detail the path through which such spanning leads to performance improvement. In our study, we assessed two important mechanisms to build the varied experience: through spanning business (user) boundaries and technology (technical applications) boundaries. We separately theorized about each type of knowledge boundary spanning and then evaluated its effect on individual performance. Our results confirmed the positive, significant contribution of both boundary-spanning activities and revealed the substantial benefits from business boundary spanning.

Finally, this study also contributes to IS literature by enhancing our understanding of IS workers' performance in IS post-implementation support. Prior research presents mixed views on the role of IT support desks. For example, Santhanam et al. [46] emphasized the effectiveness of IT help desk in transferring technical knowledge to users. However, other studies (e.g., [19]) suggested that IT help desks lack business domain expertise in resolving users' problems. Findings of our study suggest that efficient operation of IT support requires IS support personnel to understand both the local context of users' business units and the nuance of users' interactions with the technical systems.

7.3. Implications for practitioners

Findings from this study suggest useful ways to improve IS support work and its efficiency. Experience does not always lead to improved performance because the experience available to an organization or individual may be limited and difficult to interpret [30]. Our results suggest the importance of accessing to the right type of experience—varied experience accumulated from engaging with multiple business units and technical systems—in improving the performance of IS support personnel. Learning from the varied experience, IS support personnel may be better able to anticipate users' information needs. IS managers should consider assigning their employees to work with diverse user groups and to become proficient with multiple applications. Doing so allows organizations to better align knowledge management process with their employees' near-term tasks and to maximize their investments on knowledge workers.

Further, our results show that social interactions among IS workers and users play an important role in improving IS support work, suggesting useful seeding strategies for enhancing organizational support of IS. For example, an organization may identify knowledge leaders among IS workers and assign them a wide range of tasks associated with users from diverse business units or related to different technical domains. Positioning these knowledge leaders centrally in the network and champion them is an important strategy, as their accumulated knowledge may be spread onto peer workers and result in performance enhancement in the organization as a whole.

8. Concluding remarks

How to effectively cultivate and distil diverse experience in individuals continues to challenge managers. In this regard, our investigation of the IS workers and their learning in the IS support community provides insights on the two important mechanisms: to span multiple knowledge domains and to benefit from their network positions. Moreover, our data analysis confirms the positive performance effects of those two mechanisms on IS workers' productivity. In doing so, our study has responded to the call for more nuanced theorizing and investigation on experience, knowledge and productivity [3].

IS research has adopted two different views in investigating IS post-implementation phase: one focusing on technology and emphasizing the role of technical functions [16] and the other highlighting the importance of user training and user competence in promoting organizational use of IS [7]. Our field study suggests that both perspectives should be considered in supporting IS use in organizations. A recent study on post-adoptive system use revealed that end-users encountered different types of problems with the same technical application of business intelligence (BI) systems and the BI system use problem evolved over time [17]. A promising avenue to extend this research is to study the changes in knowledge workers' productivity over time and its relationship with the workers' network position. Moreover, adopting a fine-grained measure of task complexity and investigating the mediating effect of degree centrality on the closeness centrality–productivity link could offer additional, useful insights.

Finally, the IS support community in our study demonstrates a highly centralized network structure with a power law degree distribution (exponent of degree distribution $= 2.258$) [4]. This network structure suggests that a few IS workers may interact with a large number of IS users, while many network members only have sparse interaction with others. This may be due to the nature of IS support community, where IS workers are assigned to work with users for problem resolution while users do not have much interaction with each other. Future research will enrich our understanding of network-based learning and boundary spanning by examining other network structures, such as small world network structure [59].

Appendix. Computation of *Average Effort* (AvgEff) in IS support

This appendix explains how we calculate an IS worker's average effort to measure the worker's productivity. The archival records extracted from the organization's IS support ticketing database including the date and time at which each ticket was opened and closed, but this elapsed time does not equal to one's efficiency in completing a support task, as IS support personnel can be working on multiple support tickets at the same time. However, we can obtain a good estimate of the efficiency per ticket (average effort) if we keep track of the number of tasks each individual staff works on in each day. According to the algorithm developed by Graves and Mokus [21], each IT personnel's daily effort is equally divided among the open tickets (tasks) that the individual is working on. Consistent with initial estimations of efforts adopted in Boh et al. [6] and Narayanan et al. [33], we employed the following procedure to compute the efficiency of a specific task.

Step 1. Divide the daily effort (1 unit) across all the open tickets on a particular day for each individual to obtain the daily share of effort per ticket. If there were n tickets under “OPEN” status for an individual worker for a given day, then the effort for a ticket on that day was calculated by “1/n.” For example, consider the workload of an IS worker over 3 days: two tickets (A and B) open on Day 1, four tickets (A, B, C, and D) open on Day 2, and only 1 ticket (D) open on Day 3.

The effort expended on ticket A and B includes ½ unit on Day 1, ¼ unit on Day 2. Similar, the effort expended on ticket C is ¼ unit. If Day 3 is the last open day for ticket D, then the effort expended on ticket D is ¼ unit on Day 2, and 1 unit on Day 3.

Step 2. Sum up the daily effort of that ticket across all the days during which the ticket remained "OPEN" status. In the above example, the initial estimations of the total effort are: 0.75 units for ticket A and ticket B, 0.25 for ticket C, and 1.25 for ticket D.

Step 3. Standardize the initial estimation of total effort, and use the standardized measure in the regression model.

The above estimation is based on the assumption that each individual at the support center devoted one unit of effort per day on resolving all the open tickets in that day.

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